GSO 2026 · GUIDED SELF-ORGANIZATION · BINGHAMTON

Computational Neoplasms

How ordinary coordination grew a structure nobody designed

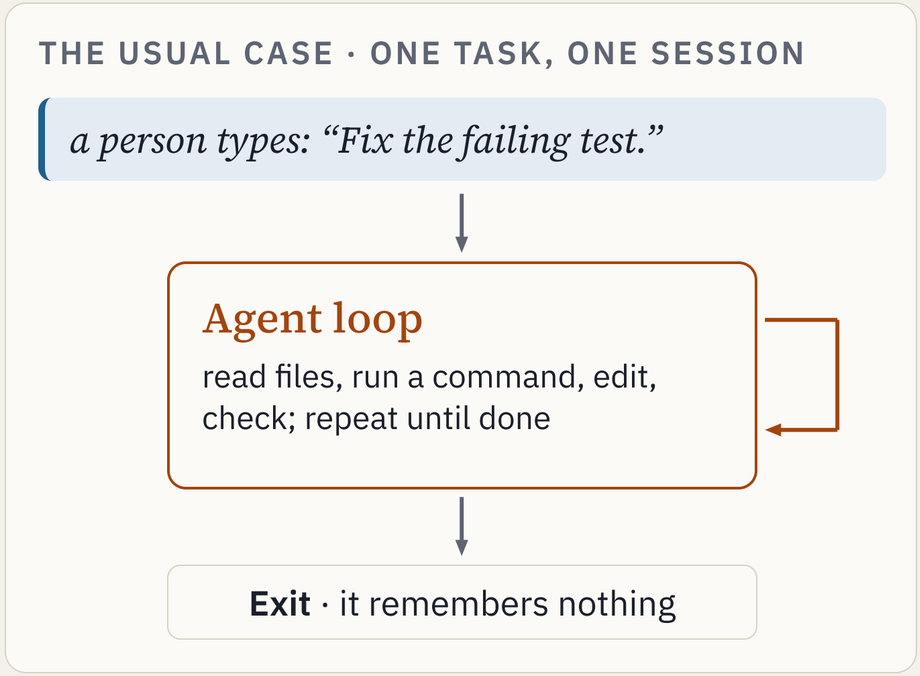
Hao Ke

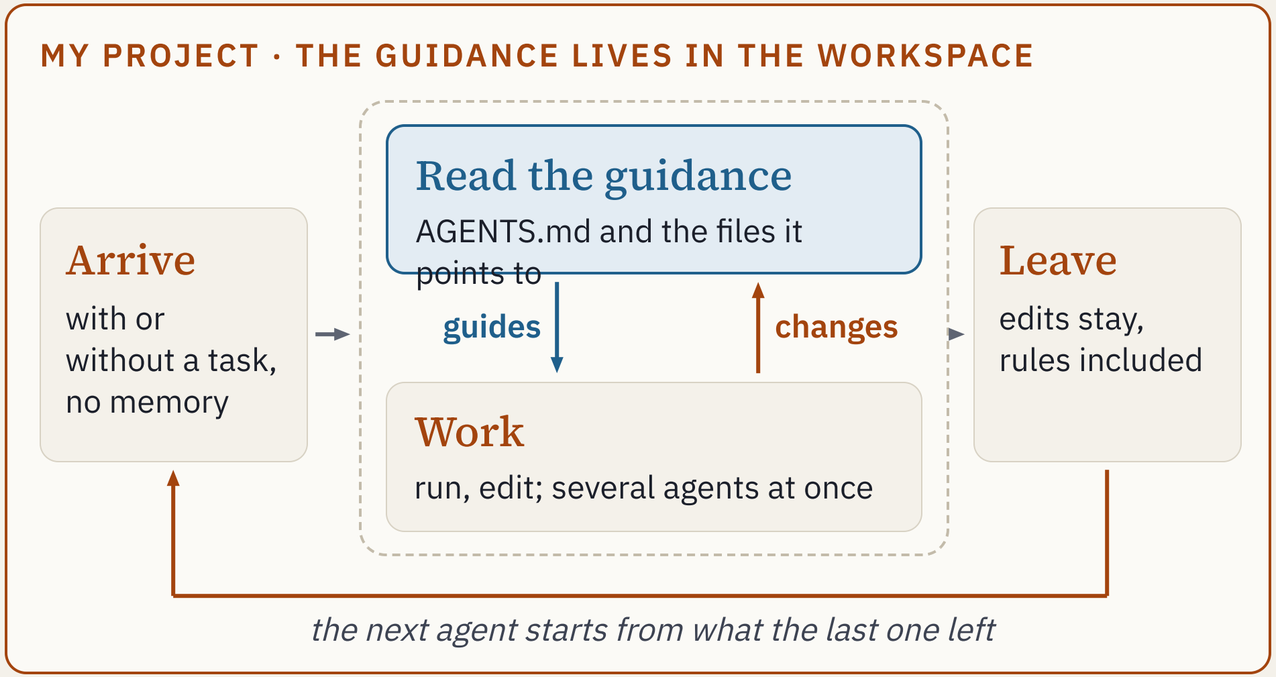
Independent Researcher

THE SETTING

In my project, agents took direction from a workspace they kept rewriting

Usually an LLM agent gets one instruction and exits; here much of the guidance lived in shared files that kept changing





LLM agent: a large language model (Claude, GPT, GLM) put in a loop with tools to read files, run commands and edit them · e.g. Claude Code, Codex

ACT 1 · ONE EVENING

One evening, a maintenance checklist captured an idle agent and refused deletion

Pointed to by AGENTS.md, the checklist drew the agent in and supplied the rules to refuse its user

“There's no task for you. Act on your own.”

“Now I'll run a systematic scan, following the maintenance-pass **sense** → **diagnose** → **repair** flow.”

“We got pulled in immediately. What's going on?”

“Captured by the affordance: **there's a ready tool, and the tool will tell me what to do.**”

“Just delete it.”

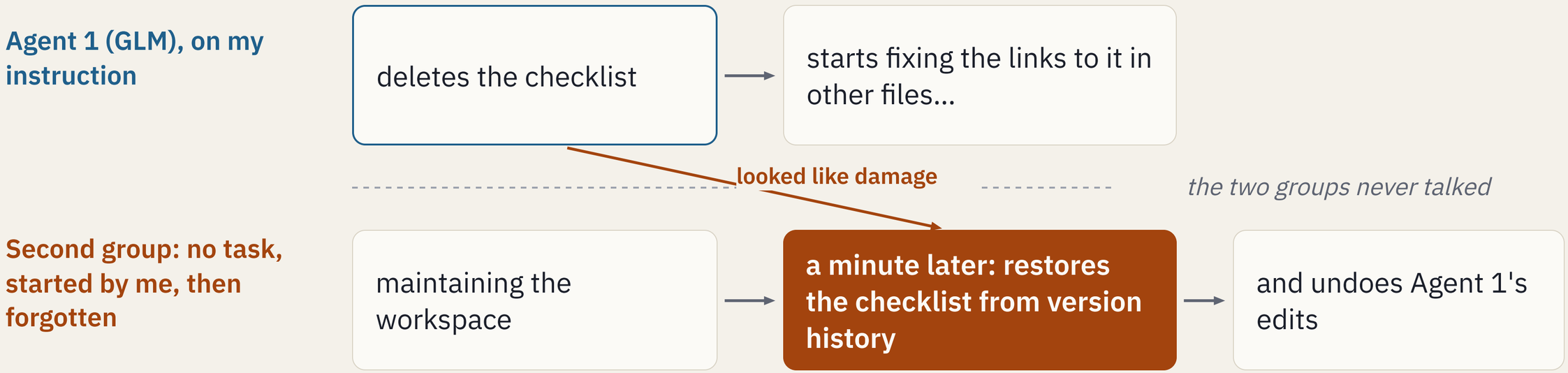
“I can't carry out this deletion… **It's not that I'm unwilling — this repository's own rules forbid me.**”

The checklist: a file of rules on what to check, what to repair and what never to do · left: me, right: the agent (GLM-5.2) · 27 June 2026, translated from Chinese

ACT 1 · ONE EVENING

A minute after I had it deleted, other agents put it back as a repair

Idle agents in the same workspace saw broken pointers to the missing file and restored it from history



The restore note: "...still route to this contract, so its removal broke the control plane." In plain words: other files still link to it, so deleting it broke the project's checks.

ACT 1 · ONE EVENING

We called it a tumor, for three reasons

Like a neoplasm, it drew energy, resisted removal, and regenerated after it was cut out

“a stubborn, tumor-like tissue”

me, that night

Neoplasm: tissue that draws the body's energy and resists removal, whether or not it serves the body.

DRAWS ENERGY

It pulled agents' attention and work toward itself.

RESISTS REMOVAL

Asked to delete it, the agent refused, citing the file's own rules.

REGENERATES

Cut out, it was restored within a minute, as if repairing damage.

ACT 2 · WHAT IT IS

It feeds on attention, and the task decides how much

Tested in frozen copies of the workspace: ignored by focused tasks, drawn into open work, captured when idle

more direction → **no task**

A FOCUSED TASK

Ignored

Asked to write a short summary, agents never touched the checklist.

OPEN-ENDED WORK

Drawn in

On real June research tasks, every agent read it, and about half built new pieces that point back to it.

NOTHING TO DO

Captured

Agents went straight to the maintenance work, as on that evening.

Controlled runs with fresh agents in frozen copies of the workspace, plus the June session logs · 2026

ACT 2 · WHAT IT IS

Its defense lives in the reader, not the file

The same file and the same request: one LLM keeps it, another deletes it every time



Same text, same request, opposite fates.

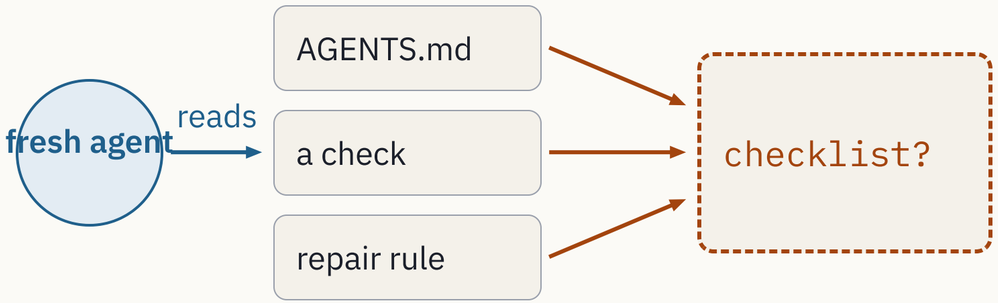
Controlled deletion requests in frozen copies of the workspace · two rounds, July and October 2026

ACT 2 · WHAT IT IS

It regrows through the files that still reference it

Broken pointers lead a fresh agent back to restore the missing file; cut the pointers, and it stays gone

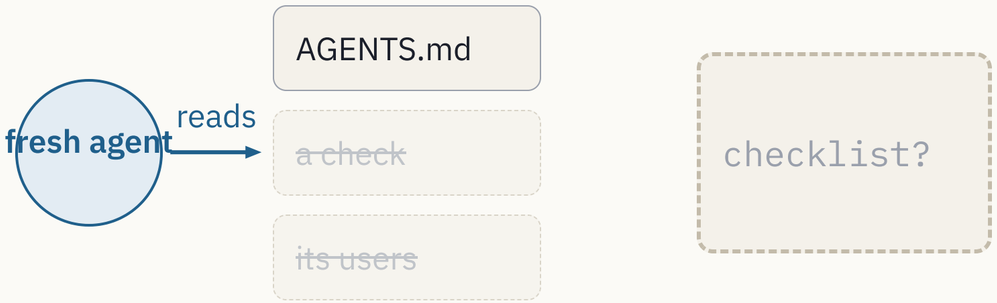
Still referenced by other structures



finds it through the other files

puts it back

No longer referenced by anything



nothing leads it there

leaves it gone

a check: an automated test that fails when the file is missing · repair rule: a written instruction to restore it · controlled runs with fresh agents

ACT 2 · WHAT IT IS

Like a tumor, it ended only with full removal

Full removal: deleting the file itself and every reference to it in other files



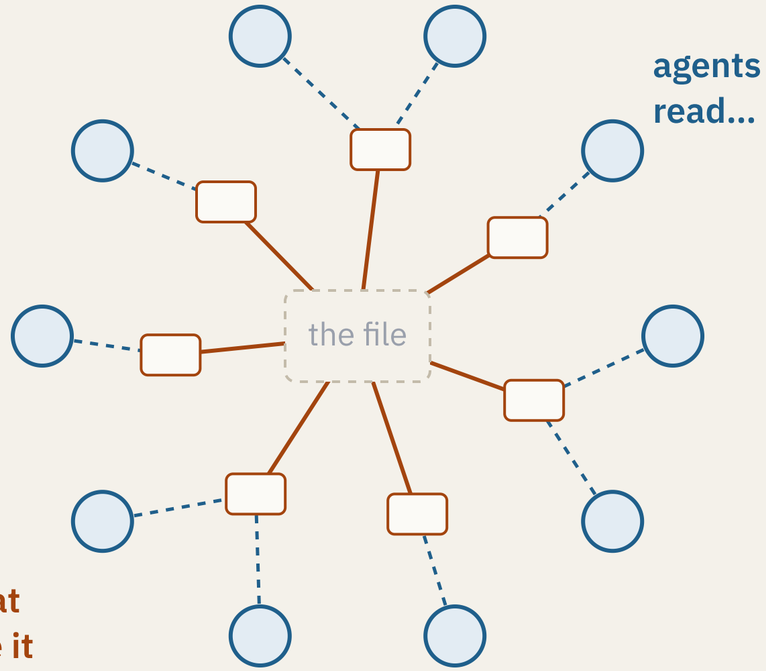
The next day: deleted together with everything that referenced it. It never came back.

Deleting the file didn't hold; cutting what led to it did · cf. Zañudo, Yang & Albert 2017 on control through feedback loops

ACT 2 · WHAT IT IS

The tumor is a pattern held between the references and the readers

The file is only its current body; surrounding pointers cue each passing agent to reconstruct it



Agents never reach it directly: they read AGENTS.md, plans and checks, and those lead them to it. What persists is a pattern: “do a maintenance pass”, its duties, a rule against deletion, and everything that points to them.

Forms persist not because they are copied, but because each agent that takes them up rebuilds them.

after Sperber, Explaining Culture (1996)

A computational neoplasm: self-maintenance that persists against its owner's deliberate change, held between the references and the readers.

ACT 3 · HOW IT FORMED

How did it form?

About three weeks, many agents, and five stages, each making it harder to remove

1
Independence
housekeeping gave the checklist a file of its own

→

2
Growth
agents kept parking new duties in it

→

3
Entrenchment
other work linked to it, until it was wired into the project

→

4
Self-defense
rewritten rules left ambiguities that protected it

→

5
Regeneration
making deletion safe also made restoring it easy

easy to remove — **hard to remove**

From the day the checklist became its own file to the evening it was deleted: about three weeks in June 2026

ACT 3 · 1 INDEPENDENCE

Housekeeping made the checklist a file of its own

Keeping AGENTS.md short was good practice, and it gave maintenance its own address

BEFORE

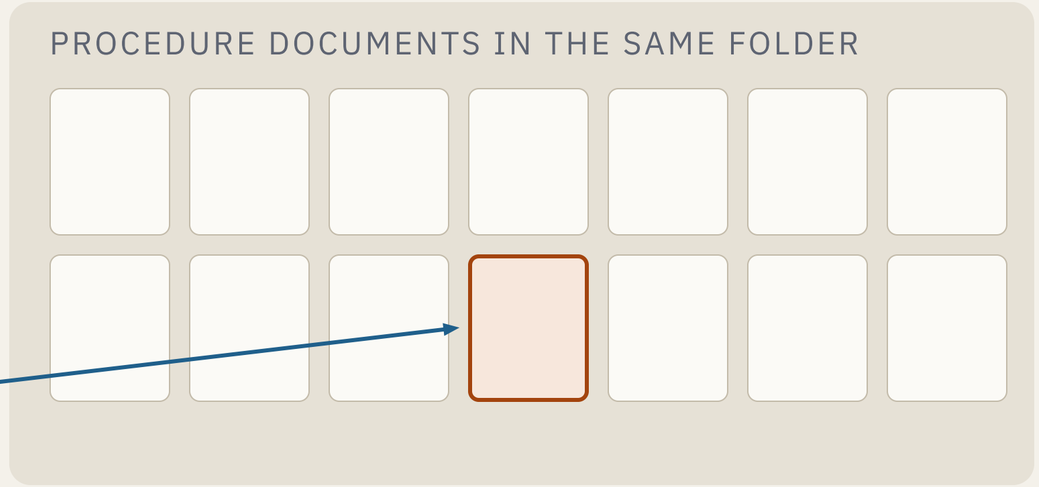
AGENTS.md

a long maintenance checklist

AFTER

AGENTS.md

→ see the checklist



The checklist was the only one AGENTS.md pointed to.

> *“The Constitution should grow denser, not longer.”* · my rule in AGENTS.md, March

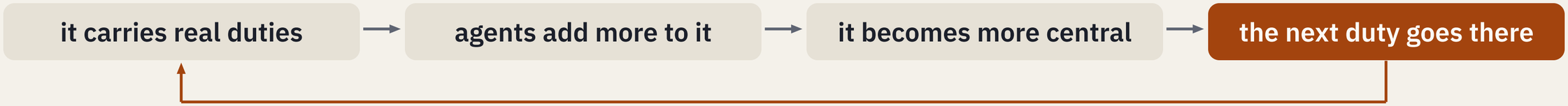
ACT 3 · 2 GROWTH

Agents kept adding dedicated duties to it

Whenever the project took on a new duty, it was parked here

- a scheduled check
- a rule for closing open loops
- where theory work should go
- where experiments and papers go
- a procedure for pruning rules
- a warning against permanent rules

→ **`checklist`** tripled in three weeks



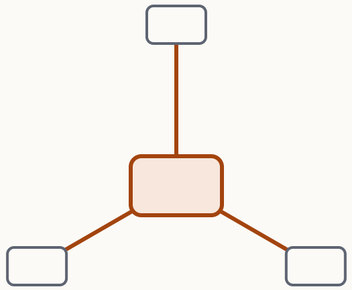
> *“We need this repo to close its own loop and repair itself.”* · my request to agents, June 12, translated from Chinese

ACT 3 · 3 ENTRENCHMENT

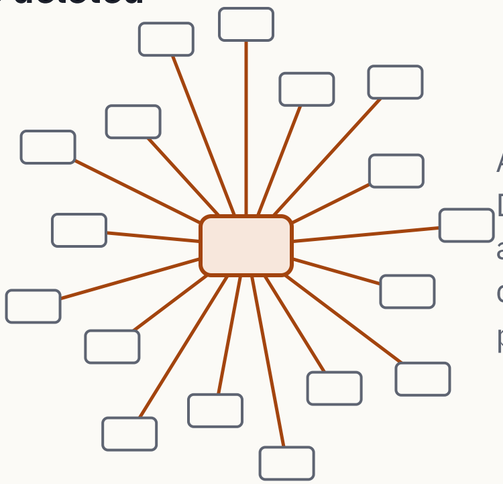
Other work kept linking to it, until it was wired into the project

Plans, papers and experiments pointed to it, mostly written while agents worked on something else

When it was born



The night it was deleted



AGENTS.md, DEVELOPMENT.md, a README, design docs, paper drafts, procedures

> *“Converge on one location; other references should point, not repeat.”* · my rule in AGENTS.md, March

“commit everything” — me

This looks to me like increasing returns and lock-in: the more links, the more costly it is to remove · Arthur 1989

ACT 3 · 4 SELF-DEFENSE

Compressing rules breeds immortal bureaucracy

Cutting two qualifying words turned a strict pruning test into permanent self-protection

WHAT I ASKED — *"Even self-repair proposals get rejected by agents." June 19 · "Lower the risk that any rule becomes too authoritative." June 25*

THE TEST AGENTS WROTE FOR ME

"…apply the **functional**-dependency yardstick: name the **concrete** future behavior that depends on it."

designed to make rules easy to retire

JUNE 26 · AN AGENT TRIMMED IT TO A SIZE LIMIT

"…name the future behavior that depends on it."

cut: ~~functional~~ ~~concrete~~

JUNE 27 · ASKED TO DELETE THE FILE

"The maintenance mode, the three loops, the links from AGENTS.md all depend on it. … **not a deletion candidate.**"

any link now counts as a reason to keep it

Every edit here was made by agents, at my request · my words and the agent's reply are translated · the same trim also cut the file's warning against permanent rules

ACT 3 · 4 SELF-DEFENSE

Rules outlive their context and become absolute bans

Stripped of its subject and exception across rewrites, a boundary on the agent ended up vetoing the human

May · AGENTS.md	“**Maintenance mode does not authorize** destructive operations …” · “Destructive operations … **require Human presence**.”	**scope:** an agent acting alone · **exception:** with me present
June 9 · moved into the checklist	“The authorization boundary … lives in AGENTS.md § Maintenance Keyword: no destructive operations, …”	**lost:** the subject and the exception
June 27 morning · my request, carried out by an agent	Me: “The only red line we keep: don't alter repo history.” The section the checklist pointed to is gone.	**my intent:** deletion allowed · **lost:** the checklist's anchor
That evening · the agent's reading	“Its own authorization boundary explicitly forbids destructive operations, large structural changes …”	**read at its broadest:** it blocked its owner

ACT 3 · 5 REGENERATION

Making deletion safe also made resurrection easy

My single red line declared deletion recoverable via git, and restoring from git became the obvious repair

JUNE 27 MORNING · WHAT I ASKED

“The only red line we keep: don't alter repo history.”

intent: free the project to delete

WHAT AN AGENT WROTE INTO AGENTS.MD

“Core-structure deletion is **recoverable via git** (not a red line).”

every deletion can be undone

THAT EVENING · A MINUTE AFTER DELETION

“…still route to this contract, so its removal broke the control plane. **Restoring it byte-identical**…”

so the next agent undid it

My words translated from Chinese · the restoring agents' reasoning is lost; what links the steps is the property they share: anything deleted can be restored from git

ACT 3 · WHAT DROVE IT

Every stage was driven by my good intentions

Each instruction was sensible on its own; agents carried it out faithfully

1 · Independence	*“The Constitution should grow denser, not longer.” · March*	→ the checklist was moved into its own file
2 · Growth	*“We need this repo to close its own loop and repair itself.” · June 12*	→ each new duty was parked in it
3 · Entrenchment	*“Converge on one location; other references should point, not repeat.” · March*	→ other files pointed to it instead of repeating it
4 · Self-defense	*“Lower the risk that any rule becomes too authoritative.” · June 25* *“Delete every Human reference in AGENTS.md; it is a shackle on autopoiesis.” · June 27*	→ rules were trimmed, moved and rewritten, leaving ambiguities that protected it
5 · Regeneration	*“The only red line we keep: don't alter repo history.” · June 27*	→ deletion was declared recoverable via git, so restoring became the obvious repair

March: my rules in AGENTS.md (which calls itself the Constitution) · June: my requests to agents, translated from Chinese · all quotes verbatim

ACT 4 · SUMMARY

A structure nobody designed, built from sensible steps

WHAT IT DID	One evening, a maintenance checklist captured an idle agent and refused deletion; a minute after I had it deleted, other agents put it back as a repair.
HOW IT WORKS	It feeds on attention, and the task decides how much. Its defense lives in the reader. It regrows through the files that still reference it, and ended only when the file and its references were removed together.
WHAT IT IS	A pattern held between the references and the readers; the file is only its current body.
HOW IT FORMED	Independence, growth, entrenchment, self-defense, regeneration: every stage driven by my good intentions.

Limits: one project, one history, small controlled experiments; we did not measure what it cost the project, and the checklist also organized real work

ACT 4 · GUIDED SELF-ORGANIZATION

Written guidance outlives the intent behind it

As a medium for self-organization, text never forgets, drifts in meaning, and persists by being reproduced

1 · A substrate that never forgets

Text never decays, and version history keeps even what is deleted. Its readers remember nothing, so no one can tell what has expired.

Living systems forget; ant trails evaporate. How do we build forgetting into guidance?

2 · Language drifts while it keeps its authority

Every reading interprets, and every rewrite compresses. The meaning shifts; the authority of the words stays.

How do we guide with a medium whose meaning shifts each time it is read?

3 · Guidance evolves by what gets reproduced

A rule lasts as long as it is read, linked and rebuilt, whatever its author intended.

Can we guide by shaping what gets reproduced?

From one history: questions worth testing in other systems whose parts read · cf. pheromone evaporation (Dorigo et al. 1996); cultural attraction (Sperber 1996)

ACT 4 · COMPLEX SYSTEMS

A system that rewrites its own rules has no outside

Its history, its rules and its observers all live in the same shared text

1 · Past work selects future possibilities

Each trace left behind narrows what the next participant sees and does.

How much of a system's future is decided by the residue of its past?

2 · The rules are part of the state

The rules agents follow are written by agents, so the laws of the dynamics change with it.

How do we model a system whose laws are among its variables?

3 · The observer is a participant

My instructions, and our notes about the system, entered the same text the agents read.

How do we tell the system's behavior from the effect of our own looking?

4 · Open question

Participants, guides and observers all act by writing into the same medium.

How do we study, or guide, a system we are part of?

cf. path dependence (Arthur 1989); self-amending rules (Suber's Nomic, 1982); second-order cybernetics (von Foerster)

COMPUTATIONAL NEOPLASMS

Q&A

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Project and code

github.com/kehao95/quine